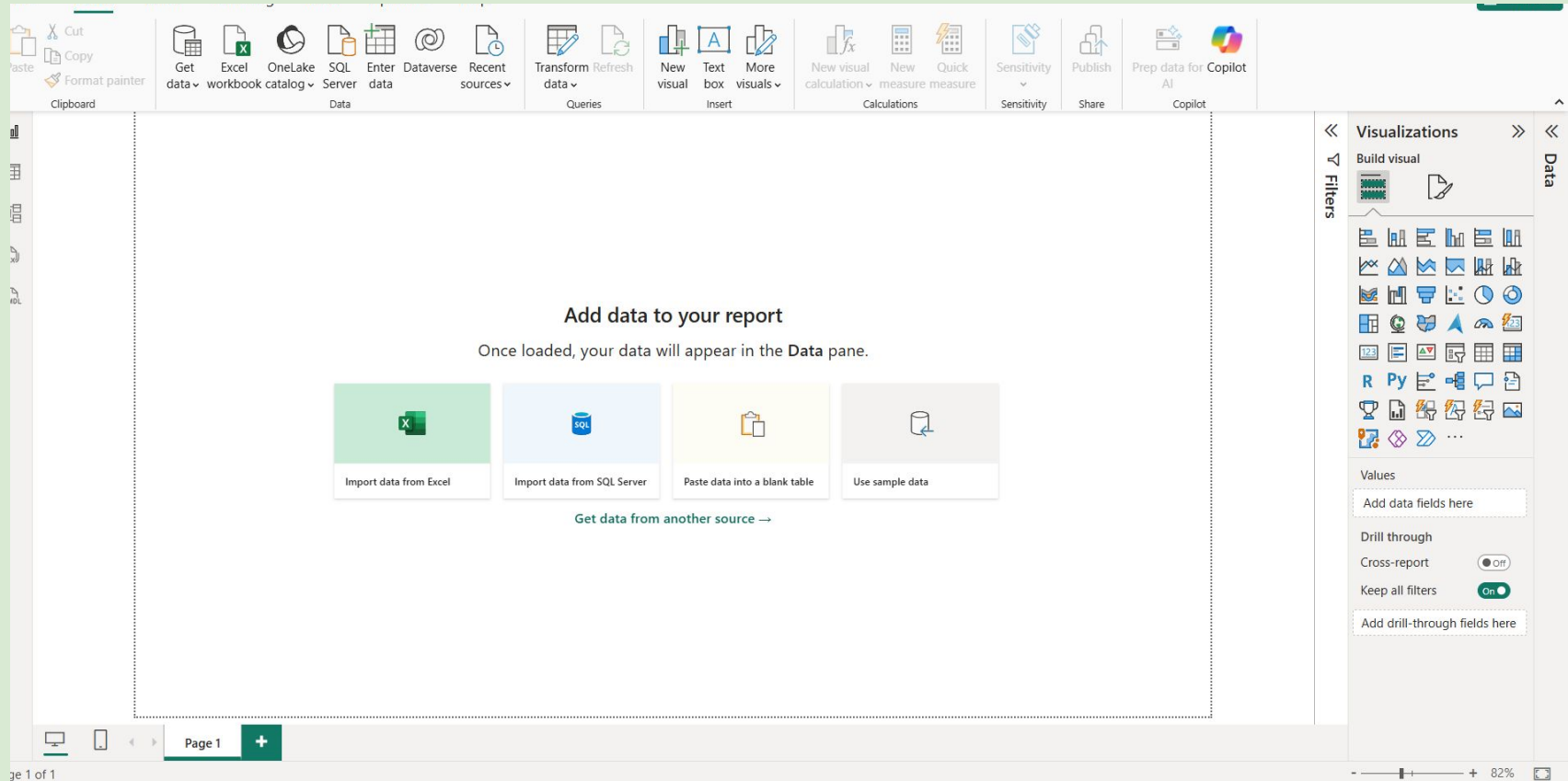


Introduction to Business Intelligence & Power BI

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IS 107 - Business Intelligence

Part 1 | Step 1: Open Power BI Desktop



Part 1 | Step 2: Load the Dataset

The screenshot shows the Power BI Desktop interface with the 'Load' dialog box open for the file 'BI_LW1_Basic_Sales_Data.csv'. The dialog displays a preview of the CSV data, which includes columns for Date, Product, Category, Region, and Sales. The 'File Origin' is set to '1252: Western European (Windows)', the 'Delimiter' is 'Comma', and 'Data Type Detection' is 'Based on first 200 rows'. The 'Load' button is highlighted in green.

Date	Product	Category	Region	Sales
01/01/2024	Product A	Electronics	East	175
02/01/2024	Product B	Furniture	South	378
03/01/2024	Product A	Electronics	North	842
04/01/2024	Product E	Furniture	North	582
05/01/2024	Product A	Electronics	South	245
06/01/2024	Product B	Furniture	North	667
07/01/2024	Product E	Furniture	West	353
08/01/2024	Product B	Furniture	East	609
09/01/2024	Product A	Electronics	West	313
10/01/2024	Product C	Office Supplies	South	434
11/01/2024	Product B	Furniture	North	494
12/01/2024	Product A	Electronics	North	539
13/01/2024	Product C	Office Supplies	East	502
14/01/2024	Product A	Electronics	West	897
15/01/2024	Product E	Furniture	West	277
16/01/2024	Product A	Electronics	East	715
17/01/2024	Product E	Furniture	South	520
18/01/2024	Product A	Electronics	South	196
19/01/2024	Product C	Office Supplies	South	231
20/01/2024	Product A	Electronics	East	539

The dialog also includes a note: 'The data in the preview has been truncated due to size limits.' and buttons for 'Extract Table Using Examples', 'Load', 'Transform Data', and 'Cancel'.

Part 1 | Step 3: Verify Data in Data View

File Home Help **Table tools** Share

Name: BI_LW1_Basic_Sales...

Manage relationships Relationships

New measure Calculations

Quick measure column

New table

Mark as date table Calendars

Structure

Table: BI_LW1_Basic_Sales_Data (400 rows)

Date	Product	Category	Region	Sales
Sunday, 21 January 2024	Product D	Electronics	East	800
Friday, 2 February 2024	Product D	Electronics	West	808
Sunday, 11 February 2024	Product D	Electronics	West	760
Thursday, 15 February 2024	Product D	Electronics	West	311
Sunday, 3 March 2024	Product D	Electronics	West	679
Saturday, 16 March 2024	Product D	Electronics	West	512
Sunday, 17 March 2024	Product D	Electronics	North	628
Tuesday, 26 March 2024	Product D	Electronics	West	647
Friday, 29 March 2024	Product D	Electronics	West	421
Saturday, 13 April 2024	Product D	Electronics	North	473
Friday, 19 April 2024	Product D	Electronics	East	706
Sunday, 5 May 2024	Product D	Electronics	North	280
Sunday, 12 May 2024	Product D	Electronics	South	835
Tuesday, 14 May 2024	Product D	Electronics	West	189
Saturday, 18 May 2024	Product D	Electronics	East	486
Saturday, 25 May 2024	Product D	Electronics	North	770
Sunday, 26 May 2024	Product D	Electronics	South	740
Tuesday, 4 June 2024	Product D	Electronics	West	830
Thursday, 6 June 2024	Product D	Electronics	North	711
Saturday, 8 June 2024	Product D	Electronics	East	743
Sunday, 9 June 2024	Product D	Electronics	West	602
Sunday, 16 June 2024	Product D	Electronics	North	775
Monday, 17 June 2024	Product D	Electronics	South	574
Tuesday, 18 June 2024	Product D	Electronics	West	656
Monday, 24 June 2024	Product D	Electronics	East	833
Tuesday, 25 June 2024	Product D	Electronics	West	777
Thursday, 27 June 2024	Product D	Electronics	East	610
Thursday, 4 July 2024	Product D	Electronics	South	860

Data

Search

BI_LW1_Basic_Sales_Data

Part 1 | Step 3: Verify Data in Data View

1.Are all columns visible?

ANSWER: Yes columns are all visible.

2. Is “Date” formatted as Date?

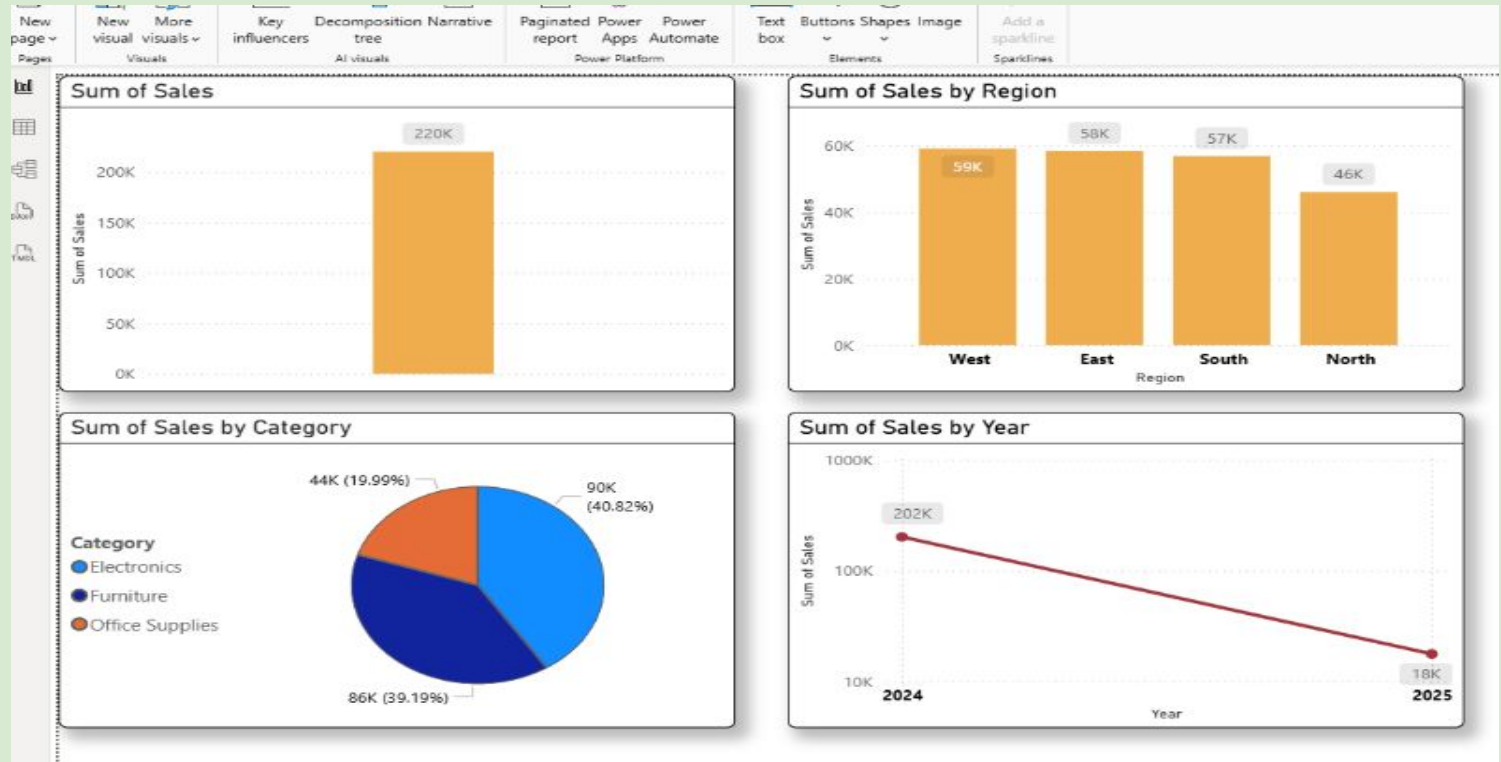
ANSWER: Yes the date was formatted as a Date.

3. Is “Sales” formatted as Decimal Number?

ANSWER: No the sales is not formatted as Decimal Number.

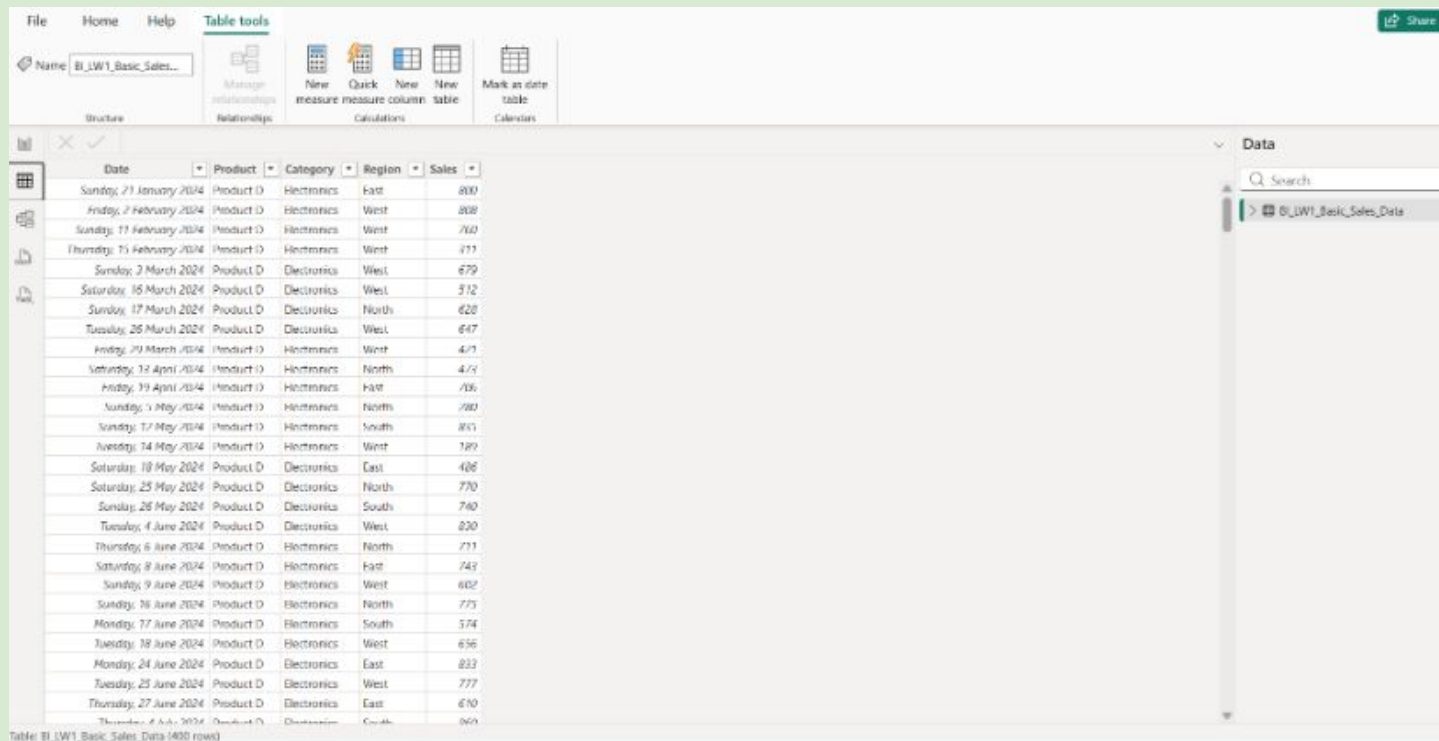
Part 2 | Exploring the Interface

Model View



Part 2 | Exploring the Interface

Data View



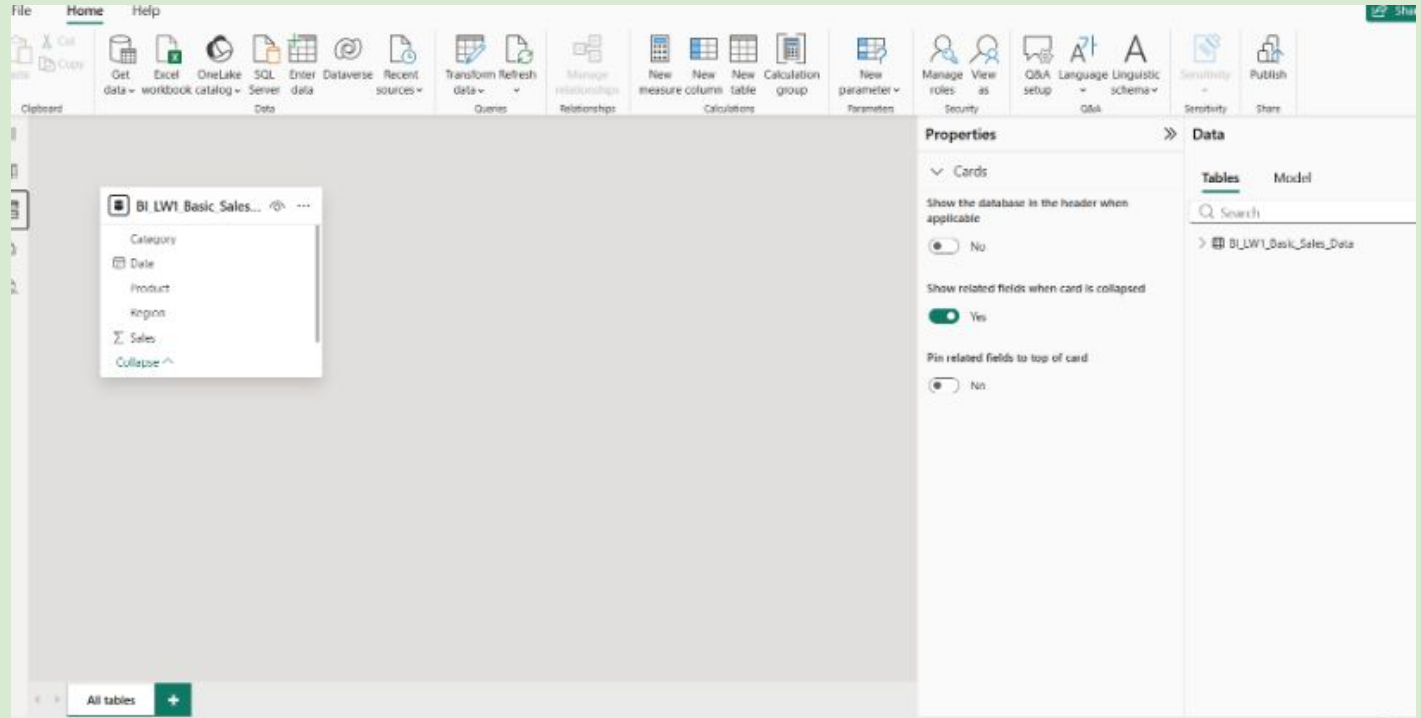
The screenshot displays the Microsoft Power BI interface. The top ribbon shows the 'Table tools' tab, which includes options for 'Structure', 'Relationships', 'Calculations', and 'Columns'. The 'Data' pane on the right shows the table 'BI_LW1_Basic_Sales_Data' selected. The main area displays a table with 40 rows of data.

Date	Product	Category	Region	Sales
Sunday, 21 January 2024	Product D	Electronics	East	800
Friday, 2 February 2024	Product D	Electronics	West	808
Sunday, 11 February 2024	Product D	Electronics	West	760
Thursday, 15 February 2024	Product D	Electronics	West	771
Sunday, 2 March 2024	Product D	Electronics	West	679
Saturday, 16 March 2024	Product D	Electronics	West	512
Sunday, 17 March 2024	Product D	Electronics	North	628
Tuesday, 26 March 2024	Product D	Electronics	West	647
Friday, 19 March 2024	Product D	Electronics	West	471
Saturday, 14 April 2024	Product D	Electronics	North	473
Friday, 19 April 2024	Product D	Electronics	East	706
Sunday, 1 May 2024	Product D	Electronics	North	780
Sunday, 17 May 2024	Product D	Electronics	North	851
Wednesday, 14 May 2024	Product D	Electronics	West	789
Saturday, 18 May 2024	Product D	Electronics	East	406
Saturday, 25 May 2024	Product D	Electronics	North	770
Sunday, 26 May 2024	Product D	Electronics	South	740
Tuesday, 4 June 2024	Product D	Electronics	West	820
Thursday, 6 June 2024	Product D	Electronics	North	771
Saturday, 8 June 2024	Product D	Electronics	East	743
Sunday, 9 June 2024	Product D	Electronics	West	602
Sunday, 16 June 2024	Product D	Electronics	North	775
Monday, 17 June 2024	Product D	Electronics	South	574
Tuesday, 18 June 2024	Product D	Electronics	West	656
Monday, 24 June 2024	Product D	Electronics	East	833
Tuesday, 25 June 2024	Product D	Electronics	West	777
Thursday, 27 June 2024	Product D	Electronics	East	610
Thursday, 4 July 2024	Product D	Electronics	East	969

Table: BI_LW1_Basic_Sales_Data (40 rows)

Part 2 | Exploring the Interface

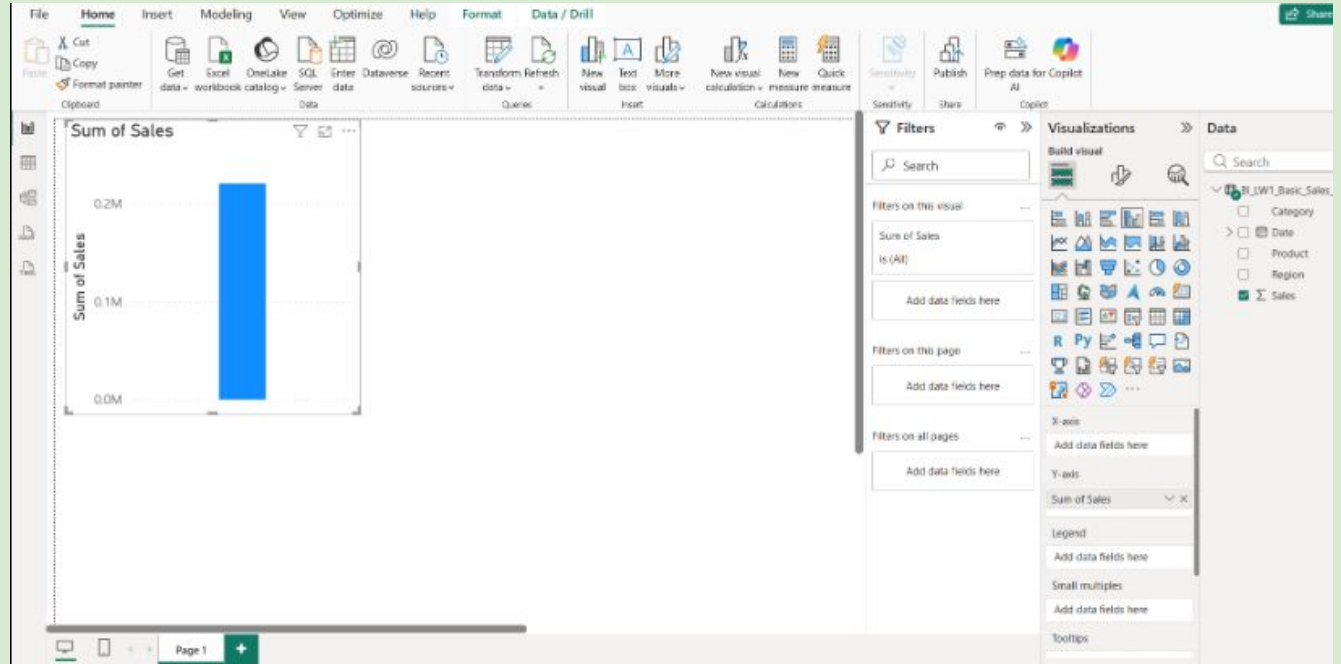
Model View



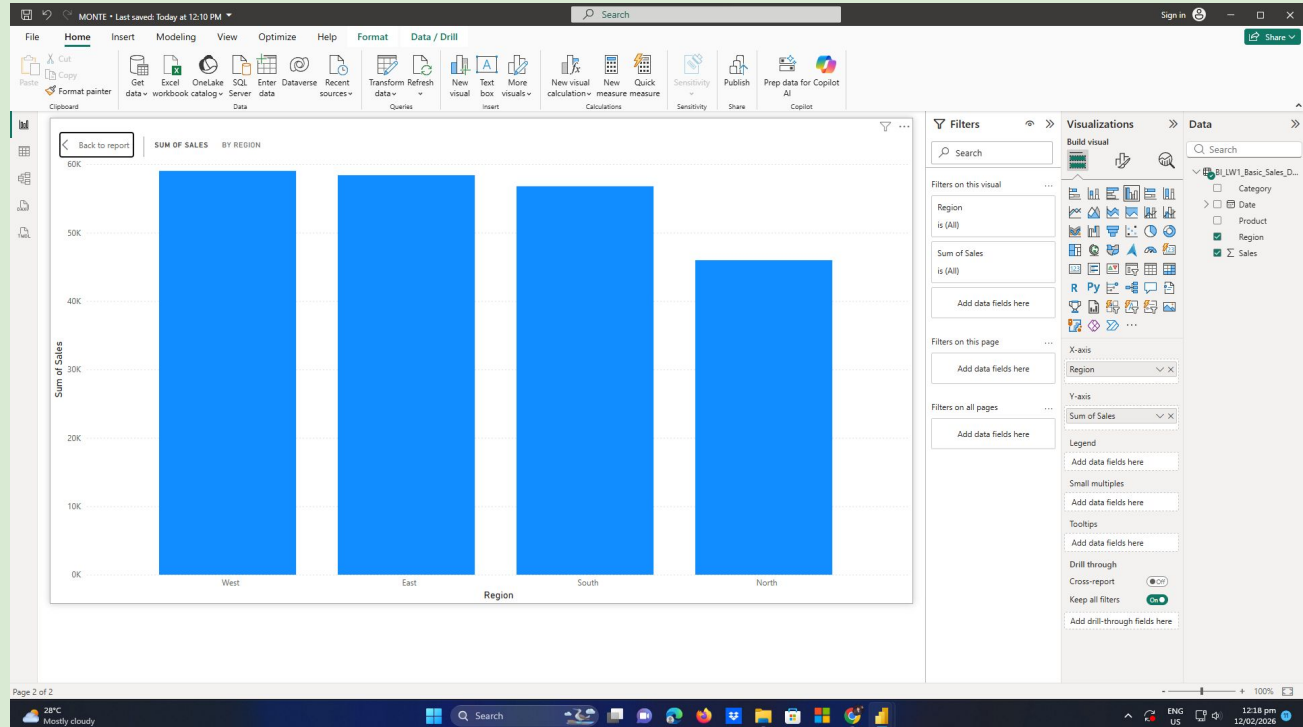
Part 3 | Step 1: Quick Visualization

1. What type of chart was created?
It was Line and Column chart.

2. What does it show?
It shows the sum of sales.

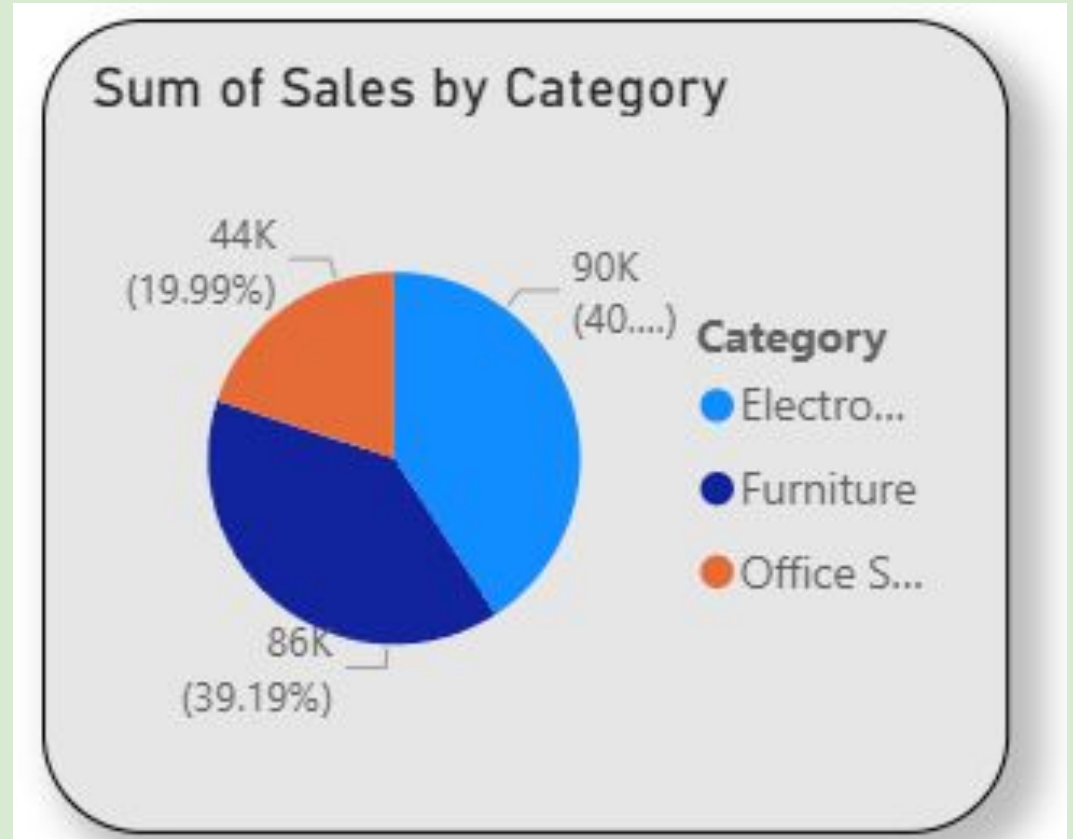


1. Which region has highest sales? The region has highest sale is West.



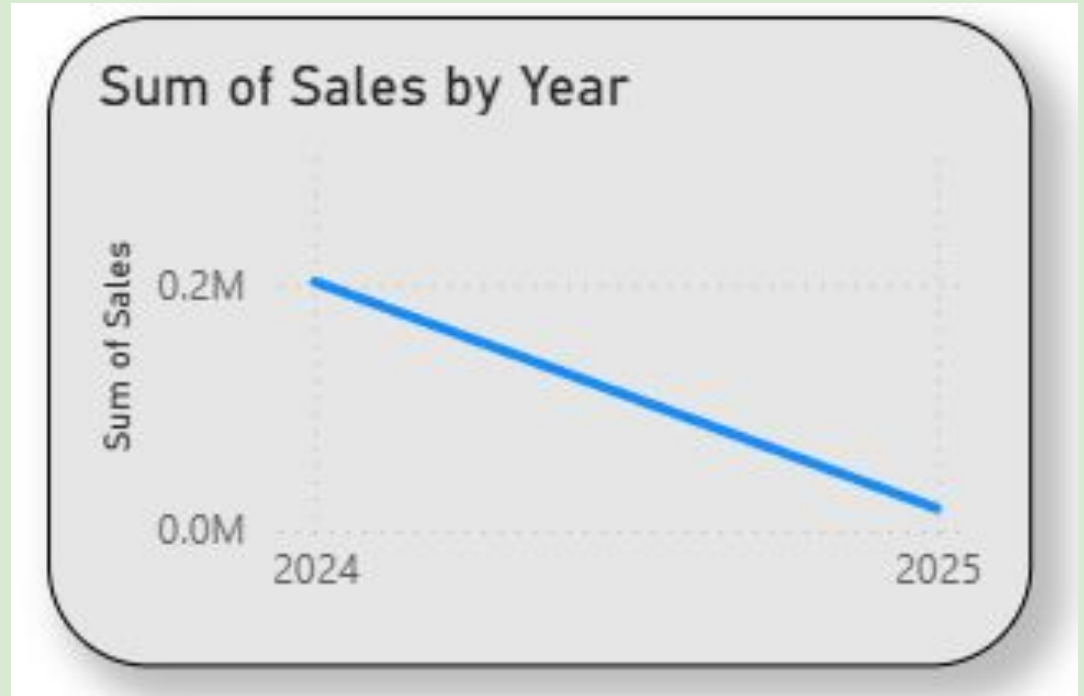
Part 3 |Step 3: Sales by Category

- 1.Which category dominates? The Electronics Category
- 2.Is the distribution balanced? The distribution balance is moderately balance between two bis category, less balance overall because office supplies is much smaller.



Part 3 | Step 4: Sales Over Time

1. Is there growth? There is NO growth
2. Any noticeable trend? There is Downward Trend



PART 4: Basic Data Insight Interpretation

1. Which region contributes most revenue? West region contributes the most revenue.
2. Which product category performs best? Electronics performs best.
3. Are sales consistent across dates? No, Sales are decreasing from 2024 to 2025.
4. What business recommendation can you suggest? Consider promotions or marketing campaigns to reverse the downward trend.

Part 4 | LABORATORY QUESTIONS

Part A – Technical Questions

1. What are the five columns in the dataset? The five columns are Date, Region, Category, Sales, Product
2. What data type is assigned to the “Sales” column? First load data was a whole number and i change it into Decimal Number
3. Which Power BI view allows you to see raw data? It is Data View
4. What chart type is best for showing trends over time? The hart type best for showing is Line Chart
5. What aggregation is automatically applied to Sales? Sum like (Sum of Sales)

Part 4 | LABORATORY QUESTIONS

Part B – Analytical Questions

6. Which region has the highest total sales? The highest region is West
7. Which category has the lowest performance? The lowest category performance was the Office Supplies
8. Are sales increasing, decreasing, or stable? the sale Decreasing
9. If you were a manager, which region would you prioritize? The North, because it has the lowest sales and needs improvement.
10. Provide one actionable recommendation based on the data. Launch a targeted marketing campaign in the North region while expanding the high-performing Electronics category to increase overall revenue and reverse the downward sales trend.

ENHANCEMENT SECTION

Task 1: Add a Card Visualization

1. What is the total sales amount?

ANSWER: The total sales is 220,000.



Task 2: Add Slicer

1. What happens to other visuals when you click a region?

ANSWER: it triggers Cross-Filtering.

2. Why is filtering important in BI?

ANSWER: Filtering is important because it turns a "big pile of data" into specific answers.



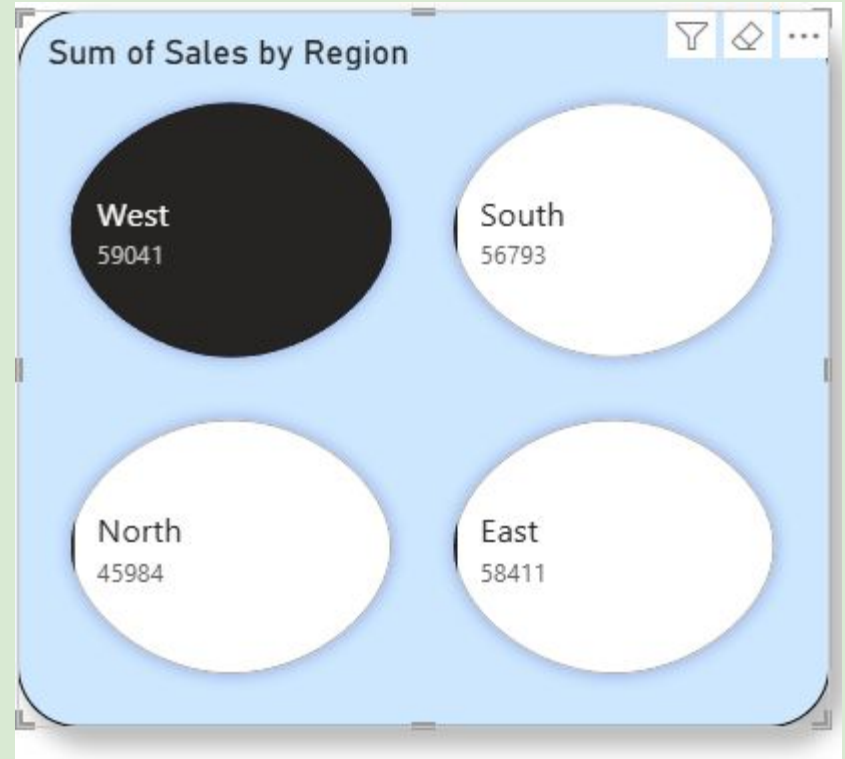
Task 3: Sort Sales

1. Does sorting improve readability?

Answer: Yes, it improve readability

2. Why?

Answer: Sorting by Sales Descending improves readability because it organizes the data into a clear ranking.



Task 4: Identify Outliers

1. Which region is significantly higher or lower?

ANSWER: Significantly the Lower is the North region and the higher is the West region.

2. What might explain that difference?

ANSWER: The North region is significantly lower because of market size or local obstacles. It likely has fewer customers, more competition, or seasonal issues like bad weather that limit sales compared to the high-performing West.